PSCAD Controller Development Guide

Unit Controller & Microgrid Controller Modeling for Hybrid AC/DC-Coupled BESS+PCS System

Reference Topology: Microgrid 1 (AC-Coupled, 5 MW/10 MWh, PCS3440) | Microgrid 2 (DC-Coupled, 2.5 MW/5 MWh, SST + DD + Super-Cap)
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1. Purpose and Scope

This document defines the PSCAD modeling methodology used to build the Unit Controller (UC) and the Microgrid Controller (MGC) for the reference hybrid microgrid shown in the system one-line diagram. The reference plant consists of two coupling architectures sharing a common 13.2 kVac bus connected to a 110 kV grid through a step-up transformer, plus a VPP/EMS interface.

- Microgrid 1 – AC-Coupled: 2 × PCS3440 (AC/DC) + 2 × BESS strings at 1332 Vdc, gas turbine (300 kW) and PV (400 kW).
- Microgrid 2 – DC-Coupled: 2 × SST (AC/DC) feeding an 800 Vdc bus with multiple DC/DC (DD) stages, BESS, and Super-Cap.
- Hierarchical control layers: SCADA Controller → Microgrid Controller → Unit Controller → Device firmware.
- Scope: PSCAD model architecture, page hierarchy, signal interfaces, control loops, tuning, and validation tests.

2. Control Hierarchy Overview

In PSCAD, the controller is replicated as a layered scheme mirroring the physical SCADA → MGC → UC chain. Each layer operates at a different time scale and is implemented on a dedicated PSCAD page module to keep the model modular, scalable, and EMT-stable.

Layer	Function	Time Scale
EMS / VPP (external)	Dispatch, market signals	minutes – hours
SCADA Controller	Mode select, setpoint routing, interlocks	100 ms – 1 s

Microgrid Controller (MGC)	P/Q dispatch, V/f regulation, islanding	10 – 100 ms
Unit Controller (UC)	PCS3440 / SST / DD inner control	50 μs – 1 ms
Device firmware (modeled)	IGBT gating, protection	<10 μs

3. Recommended PSCAD Project Structure

Build the project as a single .pscx workspace with hierarchical page modules. Each functional block becomes a reusable component, allowing duplication for the two PCS3440 units in MG1 and the two SST units in MG2.

- Top page: Grid (110 kV/13.2 kV) + 13.2 kV bus + MG1 + MG2 + EMS interface.
- MG1_AC_Coupled (page module): contains 2× PCS3440 + BESS + Gas Turbine + PV.
- MG2_DC_Coupled (page module): contains 2× SST + DC bus (800 Vdc) + DD converters + BESS + Super-Cap.
- Controllers folder: UC_PCS3440, UC_SST, UC_DD, MGC_AC, MGC_DC, SCADA_TopLevel.
- Signals: use Data Labels (global) for cross-page signals; use Import/Export tags for fast bus.
- Time step: Δt = 20–50 μs (EMT) for switching models; controller pages can run at Δt × N (multi-rate).

4. Unit Controller (UC) Modeling

4.1 Role of the Unit Controller

The Unit Controller is the device-level controller embedded in each PCS3440 (MG1) and each SST/DD (MG2). It executes the inner current loop, outer voltage/power loop, PLL, modulation, and protection. In PSCAD it is realized as a dedicated page module that takes measured voltages and currents from the converter terminals and outputs gating signals to the IGBT bridges.

4.2 PCS3440 Unit Controller (MG1 – AC-Coupled)

Build the UC as a standard grid-following / grid-forming convertible block. Follow these steps:

- Step 1 — Measurements: Insert 3-phase voltage (Vabc) and current (Iabc) meters at the PCS3440 AC terminal, plus Vdc on the 1332 Vdc DC link.
- Step 2 — abc→dq transform: Use PSCAD's 'abc to dq0' block synchronized with a SOGI-PLL (or standard PLL) locked to Vabc.
- Step 3 — PLL: Implement a SRF-PLL with bandwidth ~20–50 Hz. Output θ_pll feeds all dq transforms.
- Step 4 — Outer Loop: Build P-control and Q-control loops. P* and Q* come from the MGC. PI gains: Kp_P ≈ 0.5, Ki_P ≈ 20 (tune by step response).
- Step 5 — DC-link voltage loop: When PCS operates in grid-following charging mode, use a Vdc PI controller to generate Id*.
- Step 6 — Inner Current Loop: dq current PI controllers with cross-decoupling (ωL terms) and feed-forward of Vd/Vq.
- Step 7 — Modulation: Use SPWM or SVPWM block; carrier frequency 2–4 kHz typical for PCS3440-class IGBTs.
- Step 8 — Grid-Forming option: Add a Virtual Synchronous Machine (VSM) or droop block (P-f, Q-V) selectable by mode flag from SCADA.
- Step 9 — Protection: Add over/under voltage, over-current, and DC-link OV trip logic feeding the gating enable.

4.3 SST Unit Controller (MG2 – DC-Coupled)

The SST in MG2 is a multi-stage isolated converter (AC/DC → isolated DC/DC → 800 Vdc). In PSCAD, model it as an averaged DAB (Dual Active Bridge) plus a front-end AFE rectifier; UC has two coupled inner loops:

- Front-End AFE Loop: Regulate the intermediate DC link using a dq current loop similar to PCS3440 (grid-tied at 13.2 kVac).
- DAB Loop: Single-phase-shift (SPS) control producing phase-shift angle ϕ from a Vdc(800V) PI controller.
- Add feed-forward of load current (i_out) to reduce 800 Vdc bus transients during AI/data-center pulsed loads.
- Provide a grid-forming mode for islanded operation, where the SST regulates 800 Vdc bus voltage and AC frequency on the LV side.
- Include a fault-ride-through state machine (LVRT/HVRT) per IEEE 2800 references.

4.4 DD (DC/DC) Unit Controller

The DELTerr Ml2 BESS and Super-Cap are connected to the 800 Vdc bus via DC/DC stages (DD). Their UCs are simpler buck/boost controllers with two operating modes:

- Current-Mode (charge/discharge command from MGC): I_batt* → PI → duty cycle d.
- Voltage-Mode (DC bus support): V_dc* (800V) → outer PI → I_batt* → inner PI → d.
- Add SoC limits (e.g., 10%–90%) and current de-rating curves as lookup tables.
- For Super-Cap UC, prioritize fast transient response (bandwidth ~1 kHz) to absorb AI pulsations.

4.5 UC Interface Signals (Standard Convention)

Signal	Direction	Type	Description
P_ref, Q_ref	MGC → UC	Real	Active/Reactive power setpoint
Vdc_ref	MGC → UC	Real	DC link voltage reference (1332 V or 800 V)
Mode_cmd	SCADA → UC	Int	0 = Idle, 1 = GFL, 2 = GFM, 3 = Charge, 4 = Fault
Enable	MGC → UC	Bool	Master gating enable
P_meas, Q_meas, Vdc_meas	UC → MGC	Real	Measured feedback
Status, Faults	UC → SCADA	Word	Protection / health flags

5. Microgrid Controller (MGC) Modeling

5.1 Role of the Microgrid Controller

The MGC coordinates all UCs within a single microgrid (MG1 or MG2). It performs power dispatch, voltage and frequency regulation, mode transitions (grid-connected ↔ islanded), and resource arbitration between BESS, gas turbine, PV, and the SST/DD chain. In PSCAD it is a single page module per microgrid, executed at a slower rate (1–10 ms) than the UCs.

5.2 MGC for MG1 (AC-Coupled)

- Inputs: Bus voltage (V_bus_13.2kV), frequency (f), each PCS3440 P/Q feedback, gas turbine status, PV output, BESS SoC, EMS setpoints.
- Primary Control (droop): P-f and Q-V droop split between the two PCS3440 units (50/50 share, configurable).
- Secondary Control: PI restoration of V and f to nominal (only active in island mode) — slow loop, bandwidth ~1 Hz.

- Tertiary / Dispatch: Receives P*_plant, Q*_plant from EMS/VPP and distributes to UCs using a participation factor table.
- Mode Manager: State machine with states {GridConnected, IslandTransition, Islanded, Resync, BlackStart}.
- Resync logic: Sync-check block monitors ΔV, Δf, Δϕ vs grid before closing the PCC breaker.

5.3 MGC for MG2 (DC-Coupled)

- Inputs: 800 Vdc bus voltage, each SST output current, each DD/BESS SoC, Super-Cap voltage, building/data-center load profile.
- Primary Control (DC droop): V_dc-I droop among the two SSTs and the BESS DD for fast DC-bus sharing.
- Secondary Control: 800 Vdc bus restoration loop (PI) — keeps bus within ±1% during steady state.
- Pulsation Mitigation: Feed-forward path from load di/dt to Super-Cap UC to handle AI/data-center pulsations within microseconds.
- SoC Management: Coordinates BESS charge/discharge with PV and grid import via the SST.
- Mode Manager: {Grid-tied, Islanded-DC, Black-start-DC, Maintenance}.

5.4 MGC Block Diagram (Implementation Order in PSCAD)

- 1) Measurement aggregation page (collects all UC feedbacks via Data Labels).
- 2) Mode/State Machine block (PSCAD 'Logic' + 'Multiple Run' or custom CDF component).
- 3) Primary droop block (P-f / Q-V for AC; V-I for DC).
- 4) Secondary PI restoration block (with anti-windup).
- 5) Dispatch / participation factor block.
- 6) Synchronization & breaker control block.
- 7) Output bus to UCs (P_ref, Q_ref, Vdc_ref, Enable, Mode_cmd).

6. SCADA Controller and EMS/VPP Interface

The SCADA Controller in PSCAD is implemented as a single top-level supervisory page that issues plant-level commands to both MGCs and handles the EMS/VPP signals. For simulation, the EMS/VPP input can be a time-series file (.txt) driven by PSCAD's File Reference component.

- Drives plant-level P*, Q*, mode flags for MG1 and MG2.

- Coordinates inter-microgrid energy transfer through the 13.2 kVac bus.
- Handles alarms, trip propagation, and external utility signals (110 kV PCC).

7. Signal Flow Summary

From	To	Signals	Update Rate
EMS/VPP	SCADA	P*_plant, Q*_plant, Mode	1 s
SCADA	MGC (MG1 & MG2)	P*_mg, Q*_mg, Mode_cmd, Enable	100 ms
MGC	UC	P_ref, Q_ref, Vdc_ref, Mode, Enable	10 ms
UC	Power stage	PWM gating signals	Δt (EMT)
Power stage	UC	Vabc, Iabc, Vdc, Idc	Δt (EMT)
UC	MGC	P, Q, SoC, Status	10 ms
MGC	SCADA	Plant P, Q, V, f, Status	100 ms

8. Controller Tuning Guidelines

Loop	Bandwidth (Hz)	Phase Margin	Notes
Inner current (dq)	500 – 1000	≥ 60°	Set by switching freq / 10
DC link voltage	50 – 100	≥ 60°	Slower than current loop by 5–10×
PLL	20 – 50	≥ 45°	SOGI-PLL recommended for weak grids
P/Q outer loop	10 – 50	≥ 60°	Coordinate with MGC dispatch rate
Droop primary	1 – 10	n/a	Static gains, no dynamic comp
Secondary restoration	0.1 – 1	≥ 60°	Anti-windup required

9. Validation Test Plan in PSCAD

- T1 — Inner current loop step: 0.5 pu Id step, verify rise time < 2 ms.
- T2 — DC link disturbance: ±10% load step on 800 Vdc, verify recovery < 50 ms.
- T3 — Grid-connected P/Q dispatch: Track EMS ramp, error < 2%.

- T4 — Island transition: Open PCC breaker, verify V/f stable in < 100 ms.
- T5 — Black start (DC): SST forms 800 Vdc, BESS energizes loads in sequence.
- T6 — AI pulsation test: 1 MW square pulse at 1 kHz on MG2; verify Super-Cap absorbs ripple, BESS handles average.
- T7 — Resync: Confirm sync-check $\Delta V<3\%$, $\Delta f<0.1$ Hz, $\Delta\phi<10°$ before reclose.
- T8 — Fault ride-through: Apply 3-ph fault at 13.2 kV bus; verify LVRT per IEEE 2800.